

3.5 String methods ^

Settings

Beta features

replace(old, new)

Replace(old, new) - Returns a copy of the string with all occurrences replaced by the string new.

replace(old, new, count)

Replace(old, new, count) - Same as above, except replace(old, new) first count occurrences of old.

PythonTutor: replace() string method.

```
→ phrase = "Someday I will have three goats, six horses,

# Replace English with Spanish.
phrase = phrase.replace("one", "uno")
phrase = phrase.replace("two", "dos")
phrase = phrase.replace("three", "tres")
```

find(x)

Find(x) -- Returns the index of the first occurrence of item x in the string. If x is not found, it returns -1.

find(x, start)

Find(x, start) - Same as find(x), but begins the search at index start

find(x, start, end)

Find(x, start, end) -- Same as find(x, start), but stops the search at

rfind(x)

Rfind(x) -- Same as find(x) but searches the string in reverse, returning the string.

count(x)

Count(x) -- Returns the number of times x occurs in the string.

Figure 3.5.1: Use "in" to check if a character or substring is contained by another string.

```
if "batman" in superhero_name:  
    # Statements to execute if superhero_name contains "batman" in any position
```

Try 3.5.1: String searching example: Hangman.

The following example carries out a simple guessing game, allowing a user a number of guesses



▶ Run

```
1 word = "onomatopoeia"
2 num_guesses = 10
3
4 hidden_word = "-" * len(word)
5
6 guess = 1
7
8 while guess <= num_guesses and "-" in hidden_word:
9     print(hidden_word)
10    user_input = input(f"Enter a character (guess #{guess}): ")
11
12    if len(user_input) == 1:
13        # Count the number of times the character occurs in the word
14        num_occurrences = word.count(user_input)
15
16        # Replace the appropriate position(s) in hidden_word with the actual character
17        position = -1
18        for occurrence in range(num_occurrences):
19            position = word.find(user_input, position +
20                                1) # Find the position of the next occurrence
21            hidden_word = (hidden_word[:position] + user_input +
22                            hidden_word[position + 1:])
```

CONSOLE

Table 3.5.1: String comparisons.

Example	Expression result	Why
"Hello" == "Hello"	True	The strings are exactly identical values.
"Hello" == "Hello!"	False	The left hand string does not end with an exclamation point.
"Yankee Sierra" > "Amy Wise"	True	The first character of the left side "Y" is greater than the first character of the right side "A".
"Yankee Sierra" > "Yankee Zulu"	False	The characters of both sides match up to the first character of the second word on the left side (the value) the first character on the right side.
"seph" in "Joseph"	True	The substring "seph" can be found starting at the second character of "Joseph".
"jo" in "Joseph"	False	"jo" (with a lowercase "j") is not in "Joseph".

ASCII table

An ASCII table provides a quick lookup of ASCII values.

PARTICIPATION
ACTIVITY

3.5.1: String comparison.

Start☐ 2x speed

student_name

teacher_name

student_name > teacher_name

01234567

Ka y , _ J o

Ka y , _ A m y

75 97 121 44 32 74

75 97 121 44 32 65

= = = = = >

Captions ^

1. Each comparison uses ASCII values.
2. Values at indexes 0-4 are the same for both student_name and teacher_name.
3. "J" is greater than "A", so student_name is greater than teacher_name.

Figure 3.5.2: Identity vs. equality operators.

```

student_name = input("Enter student name: ")

if student_name is "Amy Adams":
    print("Identity operator: True")
else:
    print("Identity operator: False")

if student_name == "Amy Adams":
    print("Equality operator: True")
else:
    print("Equality operator: False")

```

Enter student r
Identity operat
Equality operat

isalnum()

Isalnum() -- Returns True if all characters in the string are lowercase letters and the numbers 0-9.

isdigit()

Isdigit() -- Returns True if all characters are the numbers 0-9.

islower()

Islower() -- Returns True if all cased characters are lowercase letters.

isupper()

isupper() -- Returns True if all cased characters are uppercase letters.

isspace()

isspace() -- Returns True if all characters are whitespace.

startswith(x)

startswith(x) -- Returns True if the string starts with x.

endswith(x)

endswith(x) -- Returns True if the string ends with x.

capitalize()

capitalize() -- Returns a copy of the string with the first character capitalized and the rest lowercased.

lower()

lower() -- Returns a copy of the string with all characters lowercased.

upper()

upper() -- Returns a copy of the string with all characters uppercased.

strip()

strip() -- Returns a copy of the string with leading and trailing white space removed.

title()

title() -- Returns a copy of the string as a title, with first letters of words capitalized.